

Interactive Space Mission Analytics Dashboard

An interactive data visualization dashboard built on historical Space Mission data.

INTRODUCTION

Space exploration has played a vital role in advancing science, technology, communication, and global connectivity. The dataset used in this project contains historical records of **4,323 space missions** conducted between **1957 and 2022**, including information on launch countries, companies, locations, mission status, rocket costs, and launch dates. It provides valuable insights into global space activities, mission success rates, launch trends, and the contributions of different countries and organizations, making it an excellent resource for data analysis and visualization.

PROBLEM STATEMENT

Space mission data is spread across large datasets containing information on launches, countries, companies, mission status, costs, and timelines, making it difficult for researchers, students, and space enthusiasts to analyse and interpret. An interactive dashboard is needed to transform this raw data into meaningful visualizations that enable quick comparison of mission performance, launch trends, success rates, and geographical distribution.

OBJECTIVES

- Develop an interactive dashboard using historical space mission data.
- Visualize space missions by country, company, launch location, and launch timeline.
- Compare mission performance using key indicators such as success rate, mission status, and rocket cost.
- Present launch trends across years, decades, months, and quarters through dynamic visualizations.
- Help users identify patterns in global space exploration, mission outcomes, and company performance using interactive filters and dashboards.

DATASET DESCRIPTION

The **All Space Missions from 1957** dataset is a comprehensive historical record of global space launches, covering missions from the beginning of the Space Age in **1957**. The dataset was compiled from **NextSpaceFlight** and includes information about launch providers, launch sites, mission dates, rockets, mission outcomes, and rocket status. It enables the analysis of trends in space exploration, launch frequency, mission success rates, and the contributions of different countries and organizations over several decades.

Source:

All space missions from 1957. Kaggle.

<https://www.kaggle.com/datasets/agirlcoding/all-space-missions-from-1957>

PROPOSED DASHBOARD FEATURES

- **KPI Cards** — Display key performance indicators including **Total Missions**, **Success Rate**, **Successful Missions**, and **Failed Missions**.
- **Filled Map** — Shows the geographical distribution of space missions by country.
- **Bar Charts** — Compare missions by **country**, **space company**, **launch location**, **launch month**, **quarter**, and **average rocket cost**.
- **Column Charts** — Display **mission status distribution** and **space missions by decade**.
- **Line Charts** — Visualize trends in **space missions by year** and the **mission timeline**.

- **Pie Chart** — Illustrates the proportion of **mission outcomes** (Success, Failure, Partial Failure, and Prelaunch Failure).
- **Scatter Plot** — Shows the relationship between **average rocket cost** and **mission success rate** across different space companies.
- **Gauge** — Displays the **Success Rate (%)** range for interactive analysis.
- **Slicers / Filters** — Enable filtering of dashboard visuals by **Country** and **Mission Status**.

EXPECTED OUTCOMES

- Develop an interactive dashboard for analysing space mission data.
- Visualize mission trends, success rates, and rocket costs.
- Compare countries, companies, and launch locations.
- Identify mission performance and historical launch patterns.
- Support quick, data-driven insights through interactive visualizations.

CONCLUSION

The **Space Mission Dashboard** transforms historical space mission data into clear and interactive visualizations, enabling users to analyse mission trends, success rates, launch activities, company performance, and geographical distribution. It provides valuable insights that support better understanding and data-driven analysis of global space exploration in a simple and user-friendly manner.

[Link](https://public.tableau.com/app/profile/khush.chawla/viz/coderscreed/SpaceMissionOverview?publish=yes&showOnboarding=true) :- <https://public.tableau.com/app/profile/khush.chawla/viz/coderscreed/SpaceMissionOverview?publish=yes&showOnboarding=true>